

▣ 10 Probability and Counting problems

Problem 1 (The Birthday Problem)

Imagine a drum of 15 balls numbered 1 to 15 with 4 balls drawn and replaced. How likely is it that the same ball will be drawn more than once?

We will imagine drawing each ball in turn and define the event M: The ball has not been drawn before. Consider the probability that there is no match with a previously drawn ball.

Ball	Note	Probability
1 st	The first ball cannot have been drawn before so $\Pr(M) = 1 = \frac{15}{15}$	$\frac{15}{15}$
2 nd	There are 14 ways to draw the second ball if we do not want to re-draw the 1 st ball. The probability of us so doing is $\frac{14}{15}$ and as the draws are independent, we have	$(\frac{15}{15})(\frac{14}{15})$
3 rd	There are 13 ways to draw the third ball if we do not want to re-draw the 1 st or 2 nd ball. Again, using independence, we have	$(\frac{15}{15})(\frac{14}{15})(\frac{13}{15})$

Similarly, after the draw of the 4th ball, $\Pr(M) = (\frac{15}{15})(\frac{14}{15})(\frac{13}{15})(\frac{12}{15}) = 0.6471 \sim 65\%$

So, if there's a likelihood of 65% that 4 balls drawn from 15 will all be unique, we can use the complement law of probability to see that about one time in three (35%), when we perform such an experiment, we will see at least one duplicate ball.

In general, if we draw k balls from n (with replacement), the chances of at least one duplicate ball being drawn is given by $1 - \frac{{}^nP_k}{n^k}$

Example A class consists of 30 students. How likely is it that at least two share a birthday?

Solution

This amounts to asking - how likely is it that if 30 balls are drawn from 365, there will be at least two matches. $n = 365, k = 30 \Rightarrow 1 - \frac{{}^nP_k}{n^k} = 0.7063$ or over 70% I.e., in every 7 out of 10 groups of 30 people (on average) you would expect at least two people in the group to share a birthday.

Exercise Show that the minimum group size needed to ensure that it was more likely than not that there will be at least two people who have the same birthdays is 24.

- ☛ *Our intuition in relation to probability will frequently lead us badly astray. If we were to ascertain the birthdates of everybody in a class of 30 students, many of us would possibly consider it somewhat remarkable if we were to find that two students had the same birthday. The calculations above show that a shared birthday is more than twice as likely as 30 unique birthdays.*

Problem 2 (A comprehensive counting problem)

Pascal is an enthusiastic reader and had recently taken delivery of a shipment of 12 novels, 3 are by JK Rowling, 5 by Agatha Christie and 4 are by John Grisham.

- (a) If the books are to be arranged on a single shelf, how many ways are there of doing so?
Answer: $12!$
- (b) How many arrangements are possible if the Rowling books form a trilogy, and these books must be together and in published order? Answer: $10!$
- (c) How many arrangements are possible if the Christie books must be together? Answer: $8! \cdot 5!$
- (d) How many ways can 4 novels be selected if
- There is no restriction? Answer: ${}^{12}C_4$
 - Only one author may be selected? Answer: ${}^3C_4 + {}^4C_4 + {}^5C_4 = 0 + {}^4C_4 + {}^5C_4$
 - Only Christie and Grisham books should be selected with two of each? Answer: ${}^4C_2 \cdot {}^5C_2$
 - Only Rowling and Christie books should be selected with at least one of each?
Answer: ${}^3C_1 \cdot {}^5C_3 + {}^3C_2 \cdot {}^5C_2 + {}^3C_3 \cdot {}^5C_1$
- (e) If Pascal elects to read three chapters from the books today, how many ways can he do so if the order in which the chapters are read should be considered and
- There is no restriction? Answer: 12^3
 - The chapters must be from different books? Answer: ${}^{12}P_3$
 - The chapters must be from different books but he starts by reading the first of the Rowling trilogy?
Answer: ${}^{11}P_2$
 - The chapters must be from different books but he starts and ends by reading a Christie book?
Answer: ${}^5P_2 \cdot {}^{10}P_1$
- (f) Having read all books Pascal must rate them. How many ways can he specify
- The top three books (i.e., Best, 2nd best and 3rd best)? Answer: ${}^{12}P_3$
 - The best three books? Answer: ${}^{12}C_3$
- (g) The 12 books are to be placed on 4 shelves. For example, one arrangement might be to have 3 books on the top shelf, none on the second, 6 on the third and 3 on the fourth. How many such arrangements are possible if
- The particular books selected for a shelf, or their order on that shelf does not matter?
Answer: ${}^{12+4-1}C_3 = {}^{15}C_3$
 - As in i) but there must be at least one book on each shelf? Answer: ${}^{8+4-1}C_3 = {}^{11}C_3$
 - Selection and arrangement do matter? Answer: $12! \cdot {}^{15}C_3$

Problems 3 and 4 below are comprehensive probability problems with particular relevance to the assignment.

Problem 3

- The outfield players of an amateur football squad consists of 9 defenders, 6 forwards and 8 midfielders but for training sessions, all can play in any position. If a training team of 10 outfield players is selected at random, how likely is it that there will be exactly 5 midfielders selected? *Ans 0.147*
- The coaching / support staff consist of 8 individuals (manager, assistant manager etc.). Each week one of these is selected at random to wash the team kit. If the team are to play five matches in the next month, how likely is it that one individual will be required to wash the kit at least twice? *Ans 0.795*
- Against their next opponents, the manner in which goals might be scored by the team can be reasonably well modelled with a Poisson distribution with a mean time between goals of 33 minutes. How likely is it that they will score twice in the first half in this game? *Ans 0.396*
- If match attendance can be modelled as being normally distributed with mean 180 and SD 13 supporters, how likely is it that there will be at least 200 supporters at the next game? *Ans 0.062*
- In competitive games, 6 substitutes are named but only 2 can be used. The manager uses both substitutes in every game and a competition is available to spectators to guess which two the manager will select (all pairs may be assumed to be equally likely). If 80 supporters participate in this competition, how likely is it that there will be more than three winners? *Ans 0.788*

Problem 4

A courier has a fleet of 11 minivans, 14 motorbikes and 12 vans to deliver various goods for their customers.

- The fuel consumption for a van delivery can be modelled as being normally distributed with mean 17.2 and standard deviation 3.6 km / litre. How likely is it that in 5 randomly considered van deliveries, fuel consumption exceeds 20 km / litre at least twice? *Ans 0.301*
- One of the courier's vehicles is involved in a minor collision in a manner than can be modelled as a Poisson process with a mean time between collisions of 1.2 years. How likely is it that there will be no more than 3 collision-free years in the next 6 years? *Ans 0.770*
- A full-time mechanic is required to keep the vehicles in good repair and after an inspection of the tyres on all vehicles, reports to the manager that there are 11 vehicles than require at least some of their tyres replaced. If all vehicles are equally likely to be in this group of 11, how likely is it that it will consist of 5 minivans? *Ans 0.124*
- A regular customer requires 2 motorbikes, once a week to simultaneously deliver important documents to their own clients. If the courier selects the bikes to make the delivery at random and if 2 of the 14 bikes are electric, how likely is it that in one year, there will be 3 or more occasions when the selected bikes will all be electric? *Ans 0.020*
- Major repairs were required by one or other of the minivans on 7 occasions in one year. If the events of requiring repairs are independent, how likely is it that the 7 will all be different minivans? *Ans 0.915*

	R code to answer problem 3		R code to answer problem 4
a	<code>choose(8,5)*choose(15, 5)/choose(23,10)</code>	a	<code>pbinom(1, 5, pnorm(20, 17.2, 3.6, lower=F), lower = F)</code>
b	<code>1- (choose(8,5)*factorial(5))/(8^5)</code>	b	<code>pbinom(3, 6, dpois(0, 1/1.2))</code>
c	<code>ppois(1, 45/33, lower.tail=F)</code>	c	<code>(choose(11,5)*choose(26, 6))/choose(37,11)</code>
d	<code>pnorm(200, 180, 13,lower=F)</code>	d	<code>pbinom(2, 52, 1/choose(14,2), lower = F)</code>
e	<code>pbinom(3, 80, 1/choose(6,2), lower=F)</code>	e	<code>1-(choose(11,7)*factorial(7))/(11^7)</code>